

Comprehensive Product Requirements and System Analysis: EdTech SaaS Platform for Sri Lankan G.C.E A/L Private Tuition

Vision Document and Industry Research

The Sri Lankan private tuition ecosystem, particularly geared toward the General Certificate of Education Advanced Level (G.C.E A/L), represents a massive, highly competitive, and lucrative industry. The A/L examination functions as the sole gateway to state-funded universities, creating intense academic pressure and driving nearly universal participation in supplementary private education. This ecosystem is bifurcated into individual "star" educators—who may teach thousands of students across multiple districts—and formalized tuition institutes that operate as multi-branch educational hubs¹.

Following global shifts, the delivery of this education has transitioned permanently into a complex hybrid model². Large-scale operations now involve conducting physical classes with massive student cohorts in a single auditorium, while simultaneously live-streaming the session to regional franchise centers and individual online students via platforms like Zoom or native video players⁴.

An analysis of the current workflow reveals a highly fragmented logistical nightmare. Students typically register for specific subjects under independent teachers who operate within a host institute. Courses are strictly categorized into Theory Classes (concept introduction), Revision Classes (rapid review), and Paper Classes (timed mock exams)⁶. Financial transactions involve students paying recurring monthly fees, which in turn grants them access to physical or digital lectures, as well as heavily guarded printed study materials locally referred to as "tutes"³. For online cohorts, these physical tutes must be dispatched via courier services, requiring meticulous address and logistics management³. At physical venues, attendance and payment verification are managed at the gate using barcode, RFID smart cards, or fingerprint scanners⁸. Despite these advancements, severe existing problems and manual processes plague the industry. The primary existential threat to teachers and institutes is revenue leakage through unauthorized content distribution¹. Students frequently share premium Zoom links, download unprotected recorded videos, and distribute login credentials to bypass monthly fees¹. Furthermore, the financial reconciliation process relies heavily on the manual verification of physical bank deposit slips uploaded to WhatsApp or basic web forms, which is labor-intensive, slow, and highly susceptible to forgery¹¹.

These systemic inefficiencies generate distinct pain points across all stakeholder groups. Students endure significant friction, from standing in long queues for physical attendance and

payment verification to navigating disjointed systems where communication happens on WhatsApp, video on YouTube, and payments via physical banks¹². The delayed courier delivery of physical tutes for online classes frequently leaves students without materials during live streams³. Teachers experience the demoralizing loss of intellectual property and revenue due to video piracy, alongside the logistical impossibility of tracking individual student performance in massive cohorts¹⁰. Furthermore, evaluating thousands of physical A/L structured essay papers weekly represents an overwhelming administrative burden⁵. Parents lack visibility into their child's attendance and academic performance, often having to physically visit institutes to pay fees or collect materials. Administrators and Finance Officers are crushed under the weight of reconciling thousands of bank transfer slips, managing complex percentage-based revenue-sharing models between teachers and the institute, and coordinating physical hardware like fingerprint scanners across multiple entry gates⁸.

Competitor Analysis

The proposed SaaS platform must compete against two distinct software categories: global generic Learning Management Systems (LMS) and highly specialized local Sri Lankan tuition platforms.

Global LMS Platforms

Platform	Strengths & UX Highlights	Weaknesses & Missing Features
Google Classroom	Seamless Google ecosystem integration, zero-cost entry, extremely user-friendly UX for basic document sharing and assignments.	Completely lacks native monetization, DRM protection, payment tracking, and physical gate attendance bridging. Easily sharable meeting links lead to immediate revenue loss.
Moodle / Canvas LMS	Highly customizable, robust assessment engines, scalable enterprise architecture, deep academic integrations.	Poor out-of-the-box UI/UX requiring heavy customization. Lacks integration with Sri Lankan payment gateways (PayHere, Webxpay), courier management, and local smart-card hardware.
Thinkific / Kajabi	Exceptional marketing funnels, polished	Prohibitively expensive tier pricing. Built primarily for

	consumer-grade UX, excellent video delivery and landing page builders.	asynchronous, self-paced courses rather than the hybrid recurring monthly fee model tied to live physical classes standard in Sri Lanka.
ClassDojo / Teachmint	Great parent communication loops, mobile-first design, excellent gamification for K-12 environments.	Focused on traditional schools rather than independent tuition businesses. Lacks A/L specific paper class workflows, on-screen marking for structured essays, and deep revenue-sharing mechanics.
TutorBird / Teachworks	Strong scheduling, calendar management, and individual tutor invoicing. Good UX for one-on-one tutoring.	Inadequate for massive scale (classes of 1000+ students). Missing DRM video protection, hardware gate integrations, and A/L specific assessment tools.
Udemy / Coursera	World-class video delivery, global marketplace exposure, highly optimized mobile apps for offline viewing.	Closed marketplaces. Teachers cannot control their own branding, enforce monthly recurring gate fees, or distribute localized physical materials via courier integrations.

Localized Sri Lankan Platforms

Platform	Strengths & UX Highlights	Weaknesses & Missing Features
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Klass.lk	Hyper-focused on revenue protection. Features smart payment slip analysis, custom-branded smart cards, strict device login limits, and dynamic PDF watermarking ¹ .	Interface functions predominantly as an administrative ERP rather than an engaging, gamified student learning environment. Student UX is functional but lacks modern engagement loops.
CFMS.lk	Dominates physical attendance hardware integration (Barcode, RFID, Fingerprint). Strong fee management, offline administrative capabilities, and SMS alert automation ⁸ .	Historically reliant on legacy or hybrid on-premise architectures. Markedly weaker in digital video delivery, DRM, and sophisticated online examination workflows. Utilitarian UX built for gate speed.
Codevus (thislms) / Eclass.lk	Offers secured video to prevent downloading and screen recording. Notably features on-screen paper marking which is vital for A/L structured essays ⁴ .	Premium pricing models that may exclude smaller individual teachers. Workflows bridging physical courier logistics with digital access remain somewhat disconnected.
EdusTutor / VTISL	Strong focus on live online classes, parent updates, and structured academic monitoring ¹⁵ .	Primarily operates as an institute rather than a pure SaaS platform for others to white-label. Lacks the physical gate hardware integration required for hybrid models.
Sisulka /Clazz.lk	Excellent logistics integration, offering free tute delivery and a mix of physical and online management ³ .	Proprietary systems built for specific institutes rather than scalable, multi-tenant SaaS offerings available to the broader market.

Possystem.lk / Akaza / Aspitio	Offers comprehensive school/institute management, multi-language support (Sinhala/Tamil/English), and automated timetabling ² .	Broad focus dilutes the specialized needs of A/L private tuition (e.g., specific paper class workflows, rapid monthly fee validation at high-volume physical gates).
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Stakeholders and User Role Matrix

A sophisticated Role-Based Access Control (RBAC) architecture is critical to managing the complex hierarchy of the Sri Lankan tuition model.

Stakeholder Role	Core Responsibilities and System Interactions
Super Admin	The SaaS platform owner. Manages overarching institute onboarding, global subscription billing, white-label configurations, global infrastructure monitoring, and platform-wide analytics.
Institute Owner	Operates at the tenant level. Manages multiple physical branches, oversees total revenue, negotiates revenue-sharing agreements with teachers, and monitors overall institute health and student churn.
Teacher	The primary content creator. Conducts live classes, uploads recorded videos, creates exams, and views batch analytics. Operates as an independent contractor within the institute's tenant ecosystem.
Assistant Teacher	Hired directly by the Teacher. Responsible for evaluating MCQ, structured essay, and essay papers. Requires highly restricted access limited strictly to the on-screen marking module and rubric configuration.
Branch Manager	Oversees day-to-day operations of a

	specific physical location. Manages localized class scheduling, physical room allocations, hardware scanner uptime, and local operational staff.
Cashier	Stationed at physical branch entrances. Processes cash/card payments, reconciles end-of-day drawers, issues physical smart ID cards, and overrides gate access in exceptional circumstances.
Receptionist	Handles front-line student inquiries, manages support tickets, updates KYC profiles, and coordinates complex courier dispatches for remote student study materials.
Content Creator	A DTP operator who uploads and formats PDFs, configures online MCQs, manages the digital asset library, and schedules automated SMS/Push announcements under the Teacher's direction.
Moderator	Monitors discussion forums, live chat during embedded Zoom/Meet sessions, and reviews AI chatbot escalations to maintain discipline and filter inappropriate content.
Finance Officer	Reconciles online payment gateway settlements, audits cashier transactions, validates AI-flagged manual bank slips, processes monthly teacher payouts (revenue share), and executes refunds.
Student	The primary end-user. Consumes DRM-protected content, attends physical/online classes, submits assignments, scans physical exams for upload, tracks personal progress, and engages in community forums.

Parent	Monitors the student's attendance, tracks exam marks across subjects, receives automated SMS/Push notifications for absent alerts, and processes monthly fee payments via the parent portal.
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Software Requirement Specification (SRS) & Functional Requirements

The platform is architected into discrete micro-modules to facilitate rapid development via React and FastAPI, ensuring high cohesion and low coupling.

Identity, Access, and User Management

Module	Purpose	Features	Roles	Business & Validation Rules	Dependencies
Authentication	Secure entry.	JWT Auth, OAuth, Device tracking.	All	Max 2 active devices per student ¹³ . Forced logout on 3rd device.	Redis, Auth API.
Student Mgt.	Profile handling.	KYC, ID generation, A/L stream mapping.	Admin, Student	Must provide valid NIC/Phone. Cannot duplicate NICs.	PostgreSQL .
Teacher Mgt.	Educator profiling.	Revenue share config, subject mapping.	Inst. Owner	Teacher requires active revenue % contract to	Finance Engine.

				publish courses.	
Institute Mgt.	Tenant isolation.	Branch creation, white-labeling, room limits.	Super Admin	Branches require physical coordinates . Room capacity cannot be null.	Subdomain routing.
Student ID Cards	Physical access tokens.	Dynamic QR generation, Smart Card linking.	Admin, Cashier	ID must cryptographically link to the user_id.	Hardware Scanners.

Curriculum and Academic Structuring

Module	Purpose	Features	Roles	Business & Validation Rules	Dependencies
Subjects & Courses	Academic taxonomy.	Create Streams (Bio/Math) and Subjects.	Admin, Teacher	Courses must map to a valid subject.	Database lookup.
Batches	Cohort grouping.	Assign students by exam year (e.g., 2026).	Teacher, Student	Students can only access content published to their specific batch.	Auth Service.
Theory	Foundation	Sequential	Teacher,	Content	Content

Classes	al learning.	lesson unlocking, drip content.	Student	must follow chronological syllabus progression .	Delivery.
Revision Classes	Rapid review.	High-velocity content, summary notes.	Teacher, Student	Available only in the final 12 months before the A/L exam.	Content Delivery.
Paper Classes	Assessment focus.	Timed mock exams, strict grading workflows ⁶ .	Teacher, Student	Requires integration with On-Screen Marking module.	Exam Engine.
Schedules	Time management.	Calendar UI, Zoom link generation, room allocation.	Branch Mgr	Detects overlapping physical room allocations or teacher double-booking.	Google Calendar API.

Attendance and Physical Operations

Module	Purpose	Features	Roles	Business & Validation Rules	Dependencies
QR/Card Attendance	Physical tracking.	Barcode/RFID scanning at gates ⁸ .	Cashier, Student	Denies entry (red flash) if current month fee	WebSocket API.

				is unpaid.	
Face Recog. Attendance	Biometric tracking.	Camera-based frictionless entry.	Admin, Student	Requires explicit GDPR/PDPA consent. Fails over to QR if confidence < 98%.	AWS Rekognition .
Online Attendance	Digital tracking.	Auto-tracking of Zoom watch time and VOD completion.	System	Student must maintain active window focus for >70% of duration.	Video Player API.

Financial Management

Module	Purpose	Features	Roles	Business & Validation Rules	Dependencies
Payments & Invoices	Revenue collection.	Gateway integration, manual cash receipts, automated PDF invoices.	Cashier, Student	Online payments instantly update ledger.	PayHere/Stripe API.
Installments	Payment flexibility.	Configurable payment plans for premium courses.	Finance	Sum of installments must exactly equal total	Ledger Microservice.

				course fee.	
Discounts & Scholars	Pricing overrides.	Promo codes, sibling discounts, merit waivers.	Finance, Admin	Requires multi-level approval for waivers > 50%.	Finance Engine.
Teacher Payroll	Automated payouts.	Calculates revenue share minus institute overheads/ SMS costs.	Finance, Inst. Owner	Payouts execute on the 1st of the month. Cannot process negative balances.	Bank Transfer API.
Revenue Reports	Financial visibility.	MRR tracking, churn analytics, branch-wise P&L.	Inst. Owner	Data must be immutable and audit-logged.	Analytics Engine.

Assessment, Marking, and Evaluation

Module	Purpose	Features	Roles	Business & Validation Rules	Dependencies
Exams (MCQ/Essay)	Knowledge testing.	Timed tests, automated grading, negative marking ⁵ .	Teacher, Student	MCQ locked post-submission. Essay requires PDF upload.	Database Locks.

Marks & Rankings	Performance metrics.	District/Island-wide ranking generation.	System, Student	Ranks compute only after 100% of batch papers are graded. Ties resolved by timestamp.	Celery Workers.
Leaderboards	Gamified visibility.	Public display of top performers.	Student	Opt-out available for privacy.	Caching (Redis).
Homework & Assign.	Formative testing.	File uploads, deadlines, late submission toggles.	Student, Marker	Max file size 20MB. Late submissions flagged visually.	AWS S3.

Content Delivery and DRM Security

Module	Purpose	Features	Roles	Business & Validation Rules	Dependencies
PDF Notes	Material delivery.	Secure viewer, dynamic watermarking, restricted printing ²⁰ .	Student	PDFs cannot be downloaded externally.	WebView SDK.
Recorded Videos	VOD playback.	HLS streaming, dynamic	Student	Prevents screen recording	AWS MediaConvert.

		IP/Email flashing, hotlink protection ¹⁰ .		via hardware DRM.	
Live Streaming	Real-time lectures.	Zoom/Meet embedded directly in UI.	Student, Teacher	Links obfuscated. Disconnects if duplicate session detected.	Zoom API.
Downloads (Offline)	Mobile access.	Encrypted local storage for offline viewing.	Student	AES-256 encryption. Playable only inside the proprietary app player.	React Native FS.
Bookmarks & History	UX enhancement.	Resume playback, save key video timestamps.	Student	Syncs across authorized devices.	Database.

Communication and Support Workflows

Module	Purpose	Features	Roles	Business & Validation Rules	Dependencies
Discussion Forums	Community engagement.	Threaded Q&A, upvoting, moderation.	Student, Moderator	Auto-flagging of profanity.	NLP filtering.

Announcements	Global messaging.	Rich text broadcasting to batches.	Teacher, Admin	High priority announcements trigger push notifications.	FCM.
Push/SMS/WhatsApp	External alerts.	Absent alerts, fee reminders, class cancellations ¹⁴ .	System	SMS limited to 160 chars. Parent opt-out mandatory.	Twilio/WhatsApp API.
Email	Formal comms.	Invoices, password resets, weekly summaries.	System	Must pass SPF/DKIM/D MARC checks.	AWS SES.
Calendar & Events	Scheduling visibility.	Unified view of live classes and exam deadlines.	Student	Timezones must localize to IST (Sri Lanka).	Frontend UI.
Support Tickets	Issue resolution.	Categorized ticketing (Tech, Finance, Academic).	Student, Support	SLAs enforced. Escalates to Manager after 48h.	Email SMTP.

Advanced Features and Portals

Module	Purpose	Features	Roles	Business & Validation Rules	Dependencies
Certificate	Accreditation	Auto-generated PDF	System,	Requires 80%+	PDF

s	on.	certificates upon completion.	Student	attendance and passing final exam.	Generator.
Referral System	Growth loops.	Unique affiliate links for student referrals.	Student	Credits applied only after referred student pays first month fee.	Finance Engine.
Analytics	Data visualization .	Heatmaps, attendance trends, video drop-off rates.	Teacher, Admin	Data aggregated nightly to reduce DB load.	Celery/Redis.
Parent/Teacher/Admin Portals	Role-specific UX.	Dedicated dashboards tailored to user needs.	Specific Roles	Strict RBAC enforcement on all API routes.	React Router.
Gamification/Badges	Engagement.	Points for forum answers, study streak tracking.	Student	Points non-transferable. Streaks reset at midnight local time.	Redis Counters.

AI Opportunities and Implementations

The integration of Generative AI and Machine Learning transitions the platform from a passive administrative tool into an active, intelligent educational ecosystem.

AI Module	Implementation Strategy and Business Value
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AI OCR for Bank Slips	Implementation: AWS Rekognition/Custom Vision model extracts Date, Amount, Reference, and Account details from uploaded slip images¹¹. Value: Eliminates the massive administrative bottleneck of manually verifying thousands of monthly fee transfers, saving hundreds of admin hours.
AI Chatbot & Doubt Solving	Implementation: A Retrieval-Augmented Generation (RAG) architecture using OpenAI/Gemini, strictly grounded in the specific teacher's authorized transcripts and PDF tutes. Value: Students receive 24/7 answers to academic questions without the AI hallucinating generic or culturally irrelevant information.
Voice Transcription & Summaries	Implementation: Speech-to-text models optimized for Sri Lankan "Singlish" (code-switching between Sinhala and English). Value: Automatically generates timestamped study notes, lecture summaries, and searchable video transcripts immediately post-lecture.
AI Paper & Question Gen.	Implementation: LLMs analyze past A/L papers to generate novel MCQ and structured essay questions matching the exact difficulty and style of the national exam. Value: Massively reduces the time teachers spend creating weekly paper classes.
Student Performance Prediction	Implementation: Machine learning models (XGBoost) analyze attendance drop-offs,

	late submissions, and quiz scores to predict students at risk of failing or dropping out. Value: Enables proactive intervention, reducing churn and improving overall institute pass rates.
AI Study Planner	Implementation: Students input their target university degree. AI analyzes their current marks and generates a dynamic daily revision timetable. Value: Provides highly personalized learning pathways, highlighting weak topic detection.
OCR for Essay Marking	Implementation: Advanced Vision models read handwritten Sinhala/Tamil/English essays. While full auto-grading is high-risk, it acts as a co-pilot, highlighting missing keywords against a master rubric for the human marker. Value: Accelerates the on-screen marking workflow⁵.
AI Flashcards & Notes	Implementation: NLP extraction of key entities and formulas from PDF tutes to auto-generate spaced-repetition flashcards. Value: Enhances student retention and engagement within the mobile app.

Non-Functional Requirements (NFRs)

To support the massive concurrency spikes typical of A/L exams and the stringent security required for premium content, the system architecture must adhere to strict NFRs.

Category	Requirement Specification
Performance	API response times must remain under

	200ms at the 95th percentile. Hardware attendance synchronization (gate scanners) must validate a student in under 800ms to prevent physical queues ¹² .
Security	Implementation of Encrypted Media Extensions (EME) and Hardware-Level DRM (Widevine/FairPlay) to defeat screen recorders ¹⁰ . Dynamic visual watermarking flashing the user's NIC/Email on videos and PDFs ¹⁰ . Strict RBAC.
Scalability	Must handle 50,000 concurrent users during peak exam times. Backend architecture must utilize horizontal pod autoscaling via Kubernetes, decoupling video transcoding to background Celery workers.
Availability / Reliability	Target 99.99% uptime. Multi-AZ deployment in AWS. Database replication (Master-Slave) with automated failover mechanisms.
Accessibility	Web interfaces must comply with WCAG 2.1 AA standards, ensuring contrast ratios and screen reader compatibility.
Localization	Full UI and database support for Sinhala, Tamil, and English. UTF-8 encoding is mandatory for all text inputs, SMS, and PDF generation ² .
Data Privacy	Strict compliance with the Sri Lankan Personal Data Protection Act (PDPA). Data encryption at rest (AES-256) and in transit (TLS 1.3).
Audit Logs	Immutable, append-only logs for all financial transactions, mark alterations by Assistant Teachers, and permission

	changes by Super Admins.
Backups & DR	Automated daily snapshots of the PostgreSQL database stored in geographically distinct AWS regions. RTO (Recovery Time Objective) of 4 hours; RPO (Recovery Point Objective) of 15 minutes.
Rate Limiting	Strict rate limiting via Redis on login endpoints (to prevent brute force) and video request endpoints (to prevent aggressive scraping).
File Storage & Caching	AWS S3 for all PDFs and Videos, fronted by CloudFront CDN. Redis utilized for caching leaderboard rankings, active sessions, and high-frequency API responses.
Monitoring & Logging	Datadog or Prometheus/Grafana for real-time infrastructure monitoring. ELK stack (Elasticsearch, Logstash, Kibana) for centralized application logging.

Database Design

The system relies on PostgreSQL for robust, ACID-compliant relational data management.

Core Entities and Attributes

Entity	Primary Key	Key Attributes	Foreign Keys
User	user_id	nic, name, phone, password_hash, role_id, locale	-
Institute	institute_id	name, domain, logo_url, theme_config	-
Branch	branch_id	name, address, capacity,	institute_id

		contact_number	
Course	course_id	title, description, fee_amount, stream	teacher_id, subject_id
Batch	batch_id	year (e.g., 2026), medium (Sinhala/English)	course_id
Enrollment	enroll_id	status (Active/Suspended), enrolled_date	student_id, batch_id
Session	session_id	type (Live/Physical), start_time, end_time	batch_id
Attendance	att_id	status (Present/Absent), timestamp, method	session_id, student_id
Payment	payment_id	amount, method (Card/Slip), status, month	student_id, batch_id
Exam	exam_id	title, total_marks, start_time, end_time	batch_id
Submission	sub_id	file_url, marks_awarded, feedback	exam_id, student_id, graded_by
SupportTicket	ticket_id	category, priority, status, description	user_id

Relationships, Cardinality, and ERD Explanation

The database is heavily normalized to 3NF to ensure data integrity, which is paramount for the financial ledger and academic grading components.

- **User to Enrollment (1:N):** A student can enroll in multiple batches across different teachers. The Enrollment table serves as the central junction linking students to both academic access and financial obligations.
- **Course to Batch (1:N):** A parent course (e.g., "A/L Physics") spawns multiple yearly cohorts (e.g., "2026 Theory", "2027 Revision").
- **Session to Attendance (1:N):** A single class session generates hundreds or thousands of attendance records. The Attendance and Payment tables are architected to handle high-frequency inserts without locking the core User or Course tables, allowing Cashiers to scan barcodes rapidly at the gate.
- **Exam to Submission (1:N):** Submissions link to both the Student and the specific Assistant Teacher (graded_by) to ensure accountability in the marking process.

User Journey Workflows

1. Student Registration & Buying a Course Workflow

1. **Entry:** Student accesses the institute's custom-branded subdomain.
2. **Signup/KYC:** Inputs NIC and Phone. Verifies OTP via SMS. Selects A/L exam year and Academic Stream (e.g., Biological Science).
3. **Discovery:** Browses the marketplace. AI Recommendation engine surfaces relevant subjects.
4. **Checkout:** Selects a specific Batch. Redirected to Payment Gateway (PayHere) or prompted to upload a physical bank deposit slip.
5. **Fulfillment:** If gateway used, access is instantaneous. If bank slip uploaded, the AI OCR reads the reference and amount¹¹, flagging it for final Cashier approval.
6. **Onboarding:** Student is prompted to download the Mobile App, provided a dynamically generated QR code for physical gate access, and asked for their physical address for tute courier delivery³.

2. Hybrid Attendance & Joining Live Class Workflow

1. **Physical Entry:** Student arrives at the lecture hall and scans their Smart ID on the Cashier's hardware scanner⁸.
2. **Validation:** The system queries the Payment table for the current month's dues. If cleared, the scanner emits a green light and logs the Attendance. If unpaid, it emits a red light, and the student is routed to the cashier desk.
3. **Online Entry:** An online student logs into the React Native app or Web portal and clicks "Join Live Class."
4. **Authentication:** The system verifies the user has not exceeded their active device limit. It then embeds the Zoom/Meet feed via Web SDK, deliberately hiding the direct URL to prevent unauthorized sharing.
5. **Tracking:** Webhooks monitor the student's connection state. If they maintain focus for >70% of the session, they are marked present.

3. Paper Class Exam & On-Screen Marking Workflow

1. **Creation:** The Teacher uploads a PDF of the A/L structured essay paper and configures the grading rubric.
2. **Attempt:** The Student downloads the paper, handwrites answers offline, uses the Mobile App's camera scanner to generate a compressed PDF, and submits it before the deadline.
3. **Allocation:** The system anonymizes the submissions and distributes them to Assistant Teachers (Markers) based on workload balancing.
4. **Evaluation:** The Marker uses the On-Screen Marking module on a tablet device⁵. Using a stylus, they draw ticks, annotate mistakes, and enter marks per question. The system auto-tallies the final score to eliminate arithmetic errors.
5. **Publishing:** The system aggregates all marks, calculates district and island-wide ranks, and publishes the leaderboard. Automated SMS alerts push the results to Parents.

Mobile App Requirements

The ecosystem necessitates highly optimized React Native applications, acknowledging that a vast majority of Sri Lankan students consume educational content exclusively via mobile devices, often on constrained cellular data networks.

- **Student App:** Functions as the primary learning hub. Contains the secure video player, PDF reader, assignment uploader, and community forums.
- **Teacher/Operator App:** Utilized by branch managers and teachers to view real-time class attendance statistics, manage gate queues, and utilize the device camera as a backup QR scanner for student entry²².
- **Parent App (Portal):** A streamlined, read-only interface for tracking fee payment histories, viewing attendance records, and analyzing academic performance metrics.
- **Secure Offline Mode:** Critical for students with intermittent internet. Students can download videos over Wi-Fi. The files are chunked and encrypted (AES-256) into local storage, rendering them playable *only* within the proprietary app player, thus preventing extraction to external storage³.
- **Camera Features & Document Scanner:** CamScanner-like functionality built natively into the app to auto-crop, apply high-contrast filters, and heavily compress physical handwritten answer sheets into small PDFs for seamless upload.
- **Dynamic QR Scanner:** The app renders a dynamic QR code for physical gate access that rotates its payload every 30 seconds, defeating attempts by students to screenshot and share their IDs.
- **Push Notifications:** Firebase Cloud Messaging (FCM) integration for immediate alerts regarding class reschedules, grade publishing, and payment deadlines.
- **Dark Mode:** Native support to reduce eye strain, accommodating the late-night study habits typical of A/L students.

Future Features (5-Year Horizon)

To ensure the SaaS platform remains a market leader, the roadmap must anticipate the

evolution of global EdTech capabilities.

- **Digital Whiteboard Replay:** Evolving beyond pixel-based video rendering. The platform will capture the vector coordinate strokes of a teacher's digital pen alongside synchronized audio. Students can replay the creation of complex physics equations stroke-by-stroke, utilizing a fraction of the bandwidth required for traditional HD video.
- **Student Mental Wellness Tracking:** A/L preparation is notoriously stressful. The system will integrate sentiment analysis on forum posts or optional biometric wearable data to flag severe anxiety, offering automated, anonymous counseling resources.
- **Competition Mode (Esports for Education):** Introducing real-time, multiplayer quiz battles between districts (e.g., Colombo vs. Kandy batches) for revision classes, tapping into the intense competitiveness of the A/L culture.
- **Parent Insights & ROI Dashboards:** Advanced predictive analytics showing parents exactly how their financial investment and the student's attendance correlates to projected academic percentile improvements.
- **Automated Ambient Attendance:** Transitioning from active gate scanning to passive Bluetooth Low Energy (BLE) or advanced ambient facial recognition, allowing thousands of students to walk through physical gates without stopping.

Complete Product Roadmap

Phase	Timeline	Strategic Focus	Key Deliverables
MVP	Months 1-3	Core Logistics & Monetization	User Auth, Institute/Teacher tenant setup, Payment Gateway integration, basic Zoom embedding, Smart Card hardware integration for physical gates, basic SMS alerts.
Version 1.0	Months 4-6	Content Security & Basic Exams	VOD with Widevine DRM and watermarking, PDF security limits, MCQ Exam engine, Student & Operator Mobile Apps,

			manual paper submission workflows.
Version 2.0	Months 7-9	A/L Specific Workflows	Tablet-optimized On-screen marking module ⁵ , API integrations with local Courier services for physical tute dispatch tracking ³ , Parent portal, Advanced Analytics and Leaderboards.
Version 3.0	Months 10-12	Gamification & Automation	Offline secure mobile video downloads, automated teacher payroll / revenue sharing logic, study streaks, achievement badges, and support ticketing system.
Enterprise / AI	Year 2+	Intelligent Scaling	AI bank slip OCR ¹¹ , AI lecture voice transcription, RAG-based doubt-solving chatbots, predictive dropout analytics, multi-tenant scalable architecture refinement.

Deliverables Checklist and Project Artifacts

As part of the system analysis, the following structural deliverables define the development parameters:

1. **Vision Document:** (Detailed in Section 1).
2. **Product Requirement Document (PRD):** Scopes the market need to eliminate revenue leakage while modernizing the hybrid logistics of Sri Lankan tuition. Success metrics include a 99% reduction in unauthorized video views and an 80% reduction in physical gate queue times.
3. **Software Requirement Specification (SRS):** (Detailed in Section 4).
4. **User Stories:** e.g., *As a Teacher, I want to upload a video that is immediately encrypted and watermarked, so my intellectual property cannot be pirated on Telegram.*
5. **Use Cases:** (Detailed in Section 7 Workflows).
6. **Functional Requirements:** (Detailed in Section 4).
7. **Non-functional Requirements:** (Detailed in Section 5).
8. **Database Entity List:** (Detailed in Section 6).
9. **User Role Matrix:** (Detailed in Section 3).
10. **API Module List:** Core REST endpoints required: /auth, /users, /billing, /courses, /attendance, /media, /exams, /webhooks/zoom.
11. **Microservice Suggestions:**
 - *Auth Service:* FastAPI handling JWT and device tracking.
 - *Academic Service:* PostgreSQL managing curriculum taxonomy.
 - *Financial Ledger:* Highly secure service for payments and revenue splits.
 - *Hardware Sync Service:* Node.js WebSocket server for low-latency physical gate scanner connections.
 - *Background Workers:* Celery tasks for video transcoding and mass SMS dispatch.
12. **Screen List:** Login/KYC, Dashboard, Course Catalog, Video Player, PDF Viewer, Exam Arena, Leaderboard, Payment Checkout, Support Chat.
13. **Dashboard List:** Super Admin Global MRR, Institute Branch P&L, Teacher Batch Analytics, Student Performance Radar, Parent Attendance Log.
14. **Navigation Flow:** (Detailed in Section 7).
15. **Complete Development Roadmap:** (Detailed in Section 11).
16. **Risk Analysis:**
 - *Piracy Risk:* High probability/Critical impact. Mitigated via Hardware DRM and concurrent login blocks¹⁰.
 - *High Load Spikes:* High probability/High impact. Mitigated via Kubernetes autoscaling 15 minutes prior to scheduled massive live classes.
 - *Gate Hardware Failure:* Medium probability/High impact. Mitigated by allowing the Cashier Operator App to sync offline lists for local scanning if internet drops.
17. **Estimated Development Complexity:** High. Requires deep integration across physical hardware (scanners), third-party logistics (couriers), high-security video streaming, and complex multi-tenant financial ledgers.
18. **Priority Matrix:**
 - *Must Have:* Secure VOD, Zoom embedding without link exposure, Payment Gateways, Barcode Attendance, RBAC.

- *Should Have:* On-screen marking for essays, iOS/Android Apps, AI Slip Analysis, Courier tracking.
- *Nice to Have:* AI Chatbot, Digital Whiteboard Replay, Gamification badges.

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